
AWS Shield - DDoS Protection Service

What is AWS Shield?

AWS Shield is a managed **Distributed Denial of Service (DDoS) protection** service that safeguards applications running on AWS. It defends against **volumetric attacks**, **state-exhaustion attacks**, and **application-layer DDoS attacks**.

Types of AWS Shield

Type	Description	Key Features
 AWS Shield Standard	Automatically included with AWS services	Free, protects against common & most frequent DDoS attacks
 AWS Shield Advanced	Paid, enhanced protection for critical apps	Advanced detection, 24/7 response, cost protection, threat intelligence

Shield vs WAF

Feature	AWS Shield	AWS WAF
Purpose	Protect against DDoS attacks	Filter traffic using custom rules
Layers	Layer 3 (Network) & 4 (Transport)	Layer 7 (Application)
Usage	Automatic (Standard)	Manual configuration
Price	Standard: Free Advanced: Paid	Pay-as-you-go
Example	Blocks SYN floods, UDP floods	Blocks XSS, SQLi, bots, bad IPs

How Shield Works

- 🧠 Detects attack patterns automatically using **machine learning**.
 - 🚧 Applies **traffic engineering techniques** to absorb or mitigate bad traffic.
 - ↩️ Shield Advanced reroutes malicious traffic with AWS global infrastructure.
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🔒 Shield Advanced Features

Feature	Description
🔴 24/7 DDoS Response Team (DRT)	On-demand support from AWS security experts
📊 Advanced Metrics & Logging	Enhanced CloudWatch metrics and attack diagnostics
💰 Cost Protection	Refunds extra usage caused by DDoS-related scaling
🧠 Application Layer Protection	Integrates with AWS WAF to block bad traffic
🌐 Global Threat Environment Dashboard	Visibility into threat landscape and attack trends

🏛️ Integration With Other AWS Services

AWS Shield (especially Advanced) integrates with:

- **Amazon CloudFront** (CDN)
- **Elastic Load Balancer (ALB/NLB)**
- **Amazon Route 53** (DNS)
- **AWS Global Accelerator**

- **Amazon EC2 and ECS**
 - **AWS WAF** for additional filtering
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Enabling AWS Shield

Shield Standard

- Automatically enabled for all AWS customers on:
 - CloudFront
 - Route 53
 - Global Accelerator
 - ELB (ALB/NLB)

Shield Advanced (Console Steps):

1. Go to **AWS Shield Console**
 2. Choose **Enable Advanced Protection**
 3. Select resources (e.g., CloudFront, ALB, EC2 IPs)
 4. (Optional) Configure AWS WAF
 5. Add notification and logging settings
 6. Review and **activate protection**
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Lab Idea: Simulate DDoS Traffic Detection (Read-Only)

While DDoS traffic **cannot be simulated legally**, students can:

1. View **Shield protection metrics** via **CloudWatch**
 2. Analyze **historical threat data** in **Global Threat Dashboard**
 3. Enable **Shield Advanced (trial)** and configure notifications
 4. Test integration with AWS WAF for Layer 7 filtering
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Best Practices with AWS Shield

-  Always use **CloudFront** or **ALB** in front of public apps.
 -  Enable **Shield Advanced** for business-critical systems.
 -  Combine **AWS WAF** with Shield for comprehensive protection.
 -  Use **AWS Firewall Manager** to deploy protection across accounts.
 -  Monitor attacks using **CloudWatch** and the **Shield console**.
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Pricing Summary

Shield Type		Cost	Notes
Standard	Free		Included with supported services
Advanced	~\$3,000/month per organization (base)		Includes extra features and cost protection

Summary Cheat Sheet

Service

Use For

AWS Shield Standard	Always-on, free DDoS protection for basic workloads
AWS Shield Advanced	Enhanced DDoS protection with 24/7 support, analytics, refunds
Combine with AWS WAF	For fine-grained, application-level protection
